

DAY-3

TASK-3: Understanding How Virtual Try-On systems and Recommendation Systems work:

1. Virtual Try-On:

A Virtual Try-On system is an AI and Computer Vision system that lets users see how clothes would look on their body digitally before purchasing. It takes accurate body measurements to give perfect sizes

Its goal is to show realistic clothing fit, appearance, and size before buying. Makes it convenient for the customers so they don't have to go to physical stores and also to avoid package returns due to size issues or returning simply because they don't like how the dress looks on them.

To achieve this it uses technologies like:

1. Pose detection
2. Image Processing
3. Cloth warping
4. Deep Learning
5. Image Generation (To see the final output in real time)
6. 3D body modelling

WORKFLOW:

User Image / Video: Real time video of the customer.

Body Detection & Pose Estimation: Using landmarks for pose detection

Body Measurements Extraction: Using Euclidean distance to measure the distance between the landmarks.

Segmentation: Used to separate clothes, humans and background.

Cloth Warping: Making the desired cloth to fit the body shape of the customer to see how it looks

3D or 2D Fitting: Making a 3D model of the customer body shape and apply the clothes to it and show in real time.

Final Output: Showing in real time how the clothes look on the customer

2. Recommendation System:

Recommendation systems are AI systems that analyze user behavior and preferences to suggest relevant products, content, or services.

In fashion technology, recommendation systems can suggest clothing items, outfit combinations, and even accurate sizes based on the user's body measurements and shopping behavior. These systems collect data such as browsing history, liked items, purchases, preferred colors, and styles. AI models then analyze patterns and similarities between users and products.

Recommendation systems suggest:

- clothes
- products
- sizes
- styles
- outfits

WORKFLOW:

User Data: Collects information like user preferences, clicks, purchases, sizes, and browsing history.

Behavior Analysis: Studies how the user interacts with products and identifies patterns in their activity.

Feature Extraction: Extracts important details such as color, style, category, brand, and user interests.

Similarity Matching: Finds products or users that are similar based on extracted features and behavior.

AI Recommendation Model: Uses machine learning algorithms to predict what the user is most likely to prefer.

Personalized Suggestions: Displays customized recommendations tailored specifically to the user.